REFERENCE FRAMES

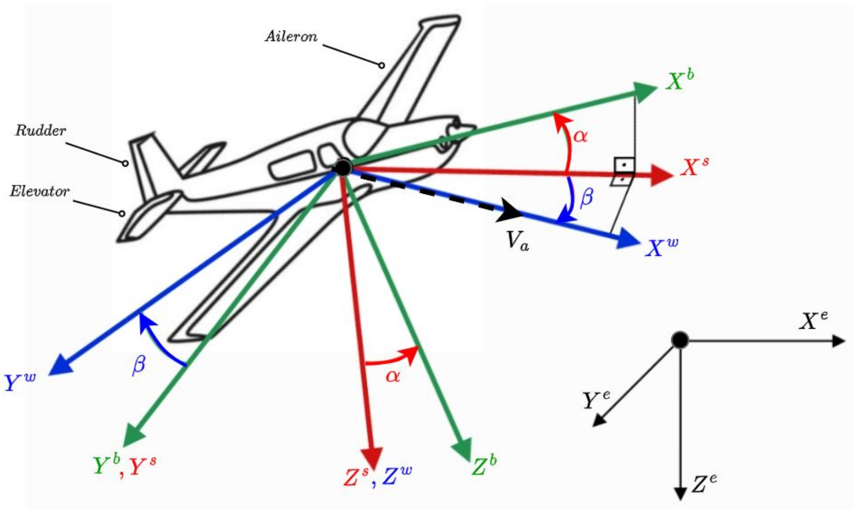


Figure 1. Reference Frames

1. Earth (Inertial) Frame - F^e

- Its a fixed reference frame attached to the Earth and generally defined by the directions North-East-Down (NED)
- Aircraft position is expressing within this frame.

2. Body Frame - F^b

- Fixed to CG of the aircraft. $\mathbf{X^b}$ towards the nose, $\mathbf{Y^b}$ towards to the right wing, $\mathbf{Z^b}$ points downward.
- IMU sensors directly measure in this frame.

3. Wind Frame - F^W

- Fixed to the aircraft and defined along the airspeed vector
- Aerodynamic forces are defined in this frame.

4. Stability Frame - F^S

- Fixed to the aircraft and rotated only by the angle of attack (α) from the body frame.
- Its a variant of wind frame and particularly useful for analysis and stability.

Rotation Matrix

Different physical phenomena are described more simply along different frame: aerodynamic forces are expressed in wind frame, equations of motion in body frame or position in earth frame. Rotation matrices are used to switch different frames. For example, earth to body and wind to body frame rotation matrices:

$$R_e^b = \begin{bmatrix} c\theta c\psi & c\theta s\psi & -s\theta \\ s\phi s\theta c\psi - c\phi s\psi & s\phi s\theta s\psi + c\phi c\psi & s\phi c\theta \\ c\phi s\theta c\psi + s\phi s\psi & c\phi s\theta s\psi - s\phi c\psi & c\phi c\theta \end{bmatrix}$$

$$R_w^b = \begin{bmatrix} \cos\beta\cos\alpha & -\sin\beta\cos\alpha & -\sin\alpha \\ \sin\beta & \cos\beta & 0 \\ \cos\beta\sin\alpha & -\sin\beta\sin\alpha & \cos\alpha \end{bmatrix}$$

AIRSPEED, ALPHA, BETA, GAMMA ANGLE DEFINITIONS

Airspeed is the aircraft's speed relative to the air mass. Its obtained by subtracting the wind speed from the ground speed. This represents the aircraft's motion relative to the air mass, and aerodynamic forces are calculated based on this speed.

$$\vec{V}_a^b = \begin{bmatrix} u - u_{wind} \\ v - v_{wind} \\ w - w_{wind} \end{bmatrix} = R_w^b \vec{V}_a^w = R_w^b \begin{bmatrix} V_a \\ 0 \\ 0 \end{bmatrix} = V_a \begin{bmatrix} \cos\alpha\cos\beta \\ \sin\beta \\ \sin\alpha\cos\beta \end{bmatrix}$$

There are several types of airspeed. Among these, IAS and TAS are the most commonly referenced. IAS for pilot operations and cockpit instruments and TAS for mostly aerodynamic calculations.

IAS is the indicated airspeed measured from the pitot tube and is unaffected by air density changes, while TAS is the actual airspeed and varies with atmospheric density.

Dynamic Pressure: $Q = \frac{1}{2}\rho_{MSL} V_{IAS}^2 = \frac{1}{2}\rho V_{TAS}^2$

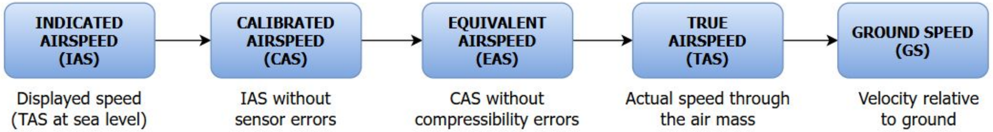


Figure 2. Types of Airspeed

Angle of attack (α) is the angle between the aircraft's longitudinal body axis and the oncoming airflow vector. It's the most important parameter for generating lift. As α increases, lift generally increases, but after a certain critical value, stall occurs.

$$\alpha = \arctan\left(\frac{w_r}{u_r}\right)$$

Side-slip angle (β) is the angle between airspeed vector and X^b-Z^b plane. A non-zero β value indicates lateral motion.

$$\beta = \arcsin\left(\frac{v_r}{\sqrt{u_r^2 + v_r^2 + w_r^2}}\right)$$

Flight path angle (γ) is the angle between the aircraft's velocity vector and the local horizontal plane, representing the actual climb or descent trajectory. Unlike pitch angle (θ) which shows where the nose points, γ shows where the aircraft is actually going.

$$\gamma = \theta - \alpha$$

FORCES AND MOMENTS

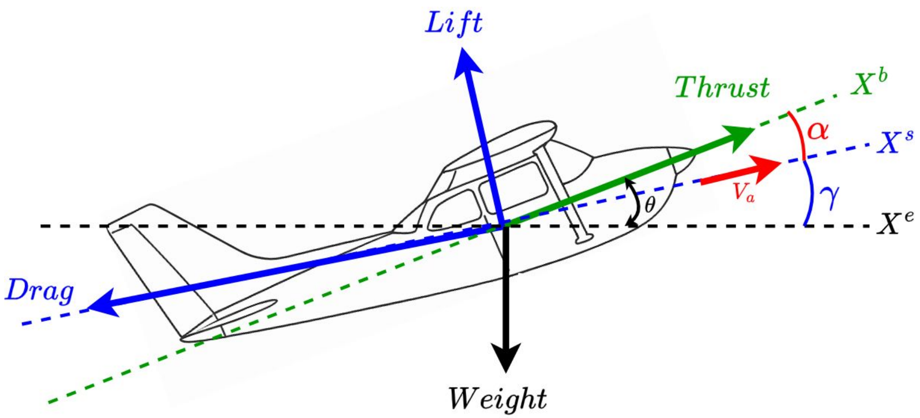


Figure 3. Forces acting on the aircraft

- $\vec{F}^b_{net} = R^b_w \vec{F}^w_{aero} + R^b_e \vec{F}^e_{grav} + \vec{F}^b_{thr}$

$$\vec{F}^w_{aero} = \begin{bmatrix} -D \\ Y \\ -L \end{bmatrix} \quad \vec{F}^e_{grav} = \begin{bmatrix} 0 \\ 0 \\ mg \end{bmatrix} \quad \vec{F}^b_{thr} = \begin{bmatrix} T \\ 0 \\ 0 \end{bmatrix}$$

$$D = QSC_D$$
$$Y = QSC_Y$$
$$L = QSC_L$$

- $\vec{M}^b_{net} = \vec{M}^b_{aero} + \vec{M}^b_{thr}$

$$\vec{M}^b_{aero} = \begin{bmatrix} l \\ m \\ n \end{bmatrix} = \begin{bmatrix} QSbC_l \\ QScC_m \\ QSbC_n \end{bmatrix}$$

S: Surface area of wing
c: Mean chord
b: Wing span

Longitudinal Aerodynamics

$$L = \frac{1}{2}\rho V_a^2 S C_L(\alpha, q, \delta_e)$$
$$D = \frac{1}{2}\rho V_a^2 S C_D(\alpha, q, \delta_e)$$
$$m = \frac{1}{2}\rho V_a^2 S c C_m(\alpha, q, \delta_e)$$

Lateral Aerodynamics

$$Y = \frac{1}{2}\rho V_a^2 S C_Y(\beta, p, r, \delta_a, \delta_r)$$
$$l = \frac{1}{2}\rho V_a^2 S b C_l(\beta, p, r, \delta_a, \delta_r)$$
$$n = \frac{1}{2}\rho V_a^2 S b C_n(\beta, p, r, \delta_a, \delta_r)$$

Dynamics:

The motion of an aircraft is determined by the sum of the forces and moments acting on it. According to Newton's 2nd law, the total forces determine the linear motion and total moments determine its rotation.

$$\vec{F}^b_{net} = m\left(\dot{\vec{V}}^b + \vec{\omega}^b \times \vec{V}^b\right)$$
$$\vec{M}^b_{net} = I\dot{\vec{\omega}}^b + \left(\vec{\omega}^b \times I\vec{\omega}^b\right)$$

AIRCRAFT STABILITY

1. Static Stability

Aircraft tendency to return to its equilibrium point after a disturbance.

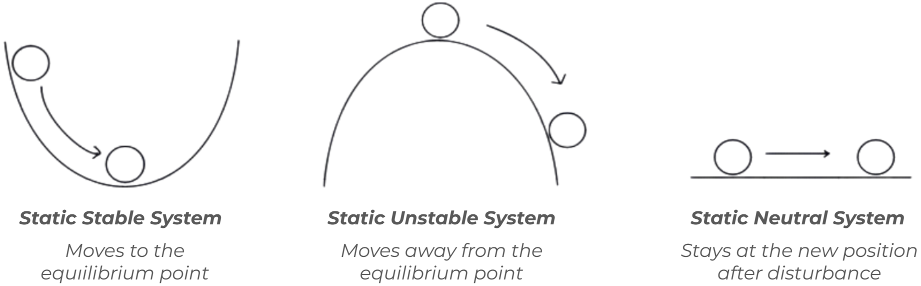


Figure 4. Static Stability Types

Static Stability Derivatives

Cm_α: Describes how the aircraft nose pitching moment changes as α increases. A negative value means the aircraft naturally pitches nose down when α increases. Stable if negative.

Cl_β: Describes how the aircraft rolls in response to sideslip. A positive value means that when the aircraft sideslips to the right, it rolls to the left. Stable if positive.

Cn_β: Describes how the aircraft yaws in response to sideslip. A positive value means that when the aircraft yaws due to a sideslip, the nose turns back into the wind. Stable if positive.

2. Dynamic Stability

- Describes how an aircraft behaves over time as it returns to equilibrium after being disturbed.
- Dynamic stability can be evaluated by linearizing the aircraft motion around its equilibrium point and analyzing the system's roots.
- **Cm_q, Cl_p, Cn_r** pitch, roll, yaw damping derivatives.

Usually they are all negative, so that a moment is produced that opposes the direction of motion to damp it.

Center of Pressure (CP)

It can be thought of as the point where aerodynamic forces act. It moves along the wing as the angle of attack changes.

Aerodynamic Center (AC)

t's a fixed point on the wing where when lift changes, the moment calculated relative to this point remains constant.

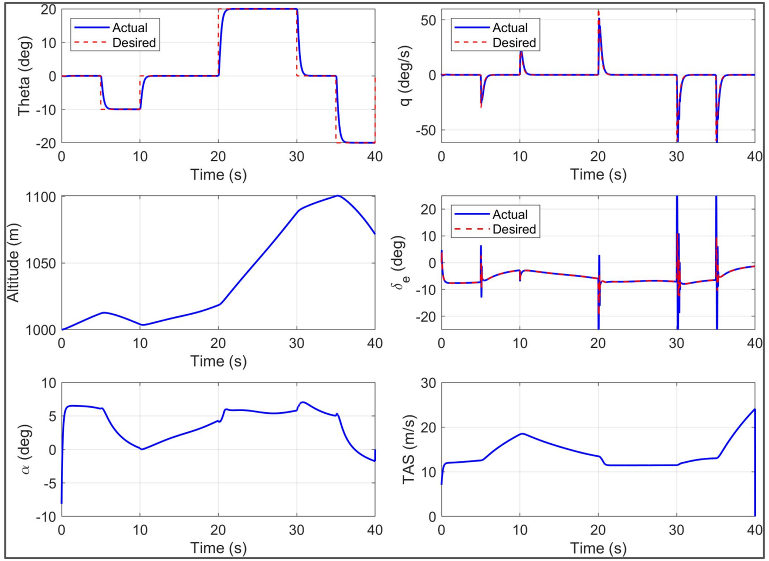
Neutral Point (NP)

It's the aerodynamic center of the entire aircraft. If CG is in front of NP, the aircraft is statically stable.

The distance between NP and CG is called **static margin** and it indicates how stable aircraft is.

INCREMENTAL NDI FOR PITCH CONTROL

Since the aircraft’s dynamics vary at different altitudes and speeds, PID control may not be sufficient under all conditions. While stable performance is achieved at a specific speed and altitude, overshoot or instability may occur at higher altitudes with the same PID gains.
In this case, Nonlinear Dynamics Inversion (NDI) can show better performance. NDI uses the model to achieve similar control performance with same gains at different conditions. But uncertainties in the model significantly affect NDI performance.
Incremental NDI (INDI), on the other hand, doesnt require the entire model. Instead, it calculates the required increase in the elevator command using the change in measured acceleration and adds this to the previous value.



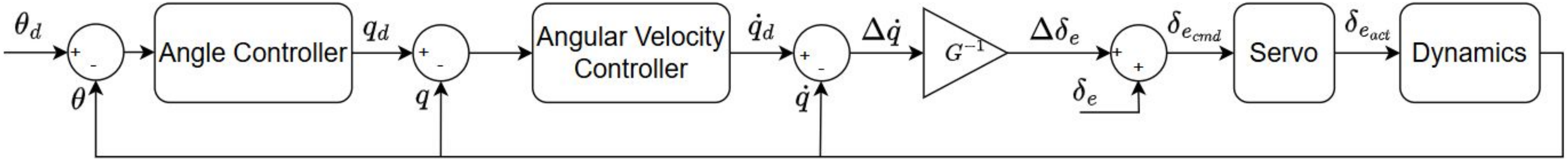


Figure 5. INDI Control Diagram

CONCLUSION

This study aims to provide an introduction to flight dynamics. Fundamental concepts such as reference frames, forces and moments and stability are examined. The study presents the mathematical foundation necessary to model and analyze the behavior of an aircraft, providing sufficient information to establish a simple simulation environment. Additionally, an INDI controller for pitching motion has been designed. For future works, different topics in flight dynamics will be examined.

References

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- Sieberling, S. & Chu, Q.P. & Mulder, J.A.. (2010). Robust Flight Control Using Incremental Nonlinear Dynamic Inversion and Angular Acceleration Prediction. Pediatric Critical Care Medicine - PEDIATR CRIT CARE MED. 33. 10.2514/1.49978.